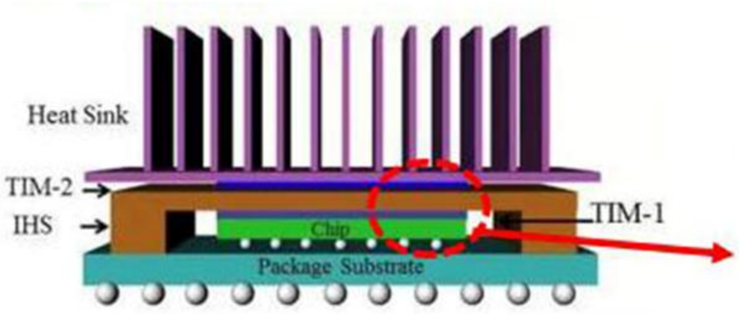
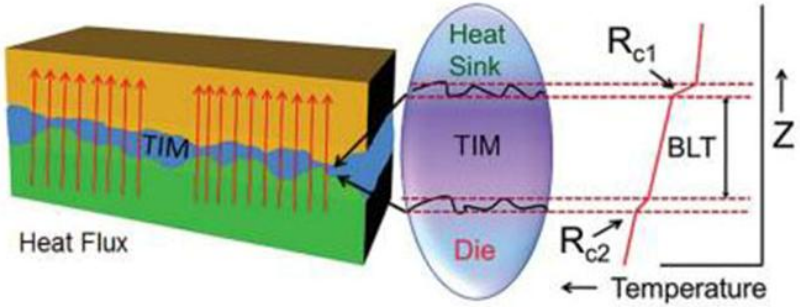
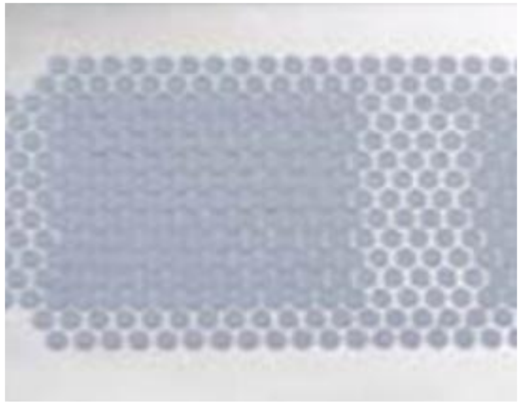
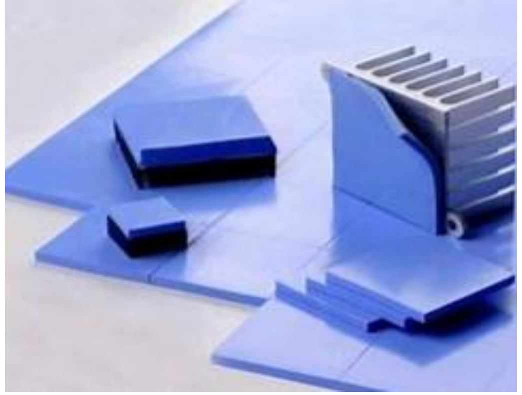
Deep-Materials Single Part Liquid Gap Filler Products

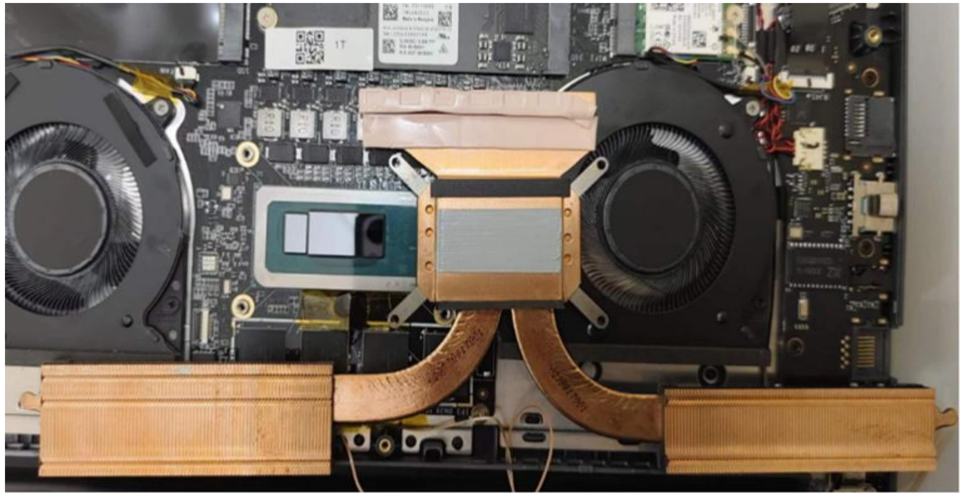
Deep Materials











Use fundamental science to make practical improvements in TIM

Deep-Materials
TIM Product Brochure

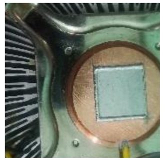
Deep Materials Mission

Use *fundamental science* to make practical improvements in Thermal Materials

CPU/GPU TIMs

Grease

- Best in class performance
- Significantly reduce pump out



Liquid Metal

- Multi-phase Liquid metal – high viscosity
- Practical to use
- Highest performance

Highest performance traditional grease & Innovative Liquid Metal grease with low resistance

Brand new category of TIM with perf. of LM but ease of grease

Gap Fillers

Gels

- Highest conductivity
- Almost no oil seepage

Gap Pads

- Almost eliminate oil seepage
- Reliable to very high temperatures

Many kinds of unique gels – e.g. ultra low oil seepage gel, low dielectric constant gels

Ultra low oil seepage & ultra low outgassing silicone pads

GAP Fillers

- For Applications in lower power/thicker TIM applications
- Do not require high pressure or mechanical support
- Typically used in consumer electronics, SSDs, Radios
- Performance measured in thermal conductivity (W/mK)

Our offerings in this space

- Liquid Gap Fillers/Gels
  - Single part (Fully cured)/Two part (in situ cure)
- Gap pads

Single Part Liquid Gap Filler

Applications:
Consumer Electronics, Automotive, Computing
SSDs, Memory Modules, Networking, Data Centers

	TCG-Gel-2000-LP	TCG-Gel-2000RW	TCG-Gel-4000	TCG-Gel-4000-LP	TCG-Gel-5000-HF	TCG-Gel-7000	TCG-Gel-10000	TCG-Gel-12000	Testing Method
Color	White	Light Yellow	Green	Yellow	Blue	Grey	Red	Grey	Visual
Minimum BLT (mm)	0.06	0.15	0.06	0.15	0.1	0.1	0.12	.12	–
Density (g/cc)	1.7	2.7	3.25	1.6	3.20	3.37	3.50	3.3	ASTM D792
Specific Heat Capacity (J/g·K)	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	ASTM E1269
Flow Rate (g/min)	40	> 200	18	40	40	12	10	10	30cc tube, 90psi air pressure
Volatility (%)	<0.01	<0.05	<0.01	<0.05	<0.01	<0.01	<0.01	<0.01	ASTM E595
Operation Temperature (°C)	-50~200	-50~200	-50~200	-50~200	-50~200	-50~200	-50~200	-50~200	–
Electrical Parameters									
Breakdown Voltage (Kv/mm)	>8.0	>8.0	>8.0	>8.0	>8.0	>8.0	>8.0	>8.0	ASTM D149
Volume Resistance (Ω.cm)	10^{14}	10^{13}	10^{13}	10^{14}	10^{13}	10^{13}	10^{13}	10^{13}	ASTM D257
Dielectric Constant (1GHz)	3.7	5.1	7.5	3.9	7.5	8.0	9.8	-	ASTM D150
Flammability rating	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	UL 94
Thermal Parameters									
Thermal Conductivity (W/m-K)	2.0	2.0	4.0	4.0	5.0	7.0	10.0	12.0	ASTM D5470

Deep Materials

Contact Us

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